

CS319: Scientific Computing

**Projects; Strings, and Files and Streams;
Vectors**

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Week 8: 6 + 8 March, 2025



Slides and examples: <https://www.niallmadden.ie/2324-CS319>

0. Outline

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Slides and examples:

<https://www.niallmadden.ie/2425-CS319>

1. Projects!

Notes for this part are at:

[https://www.niallmadden.ie/2425-CS319/
2425-CS319-Projects.pdf](https://www.niallmadden.ie/2425-CS319/2425-CS319-Projects.pdf)

2. Recall: objects

Last week we learned that

- ▶ A **class** is a general form of data type that we can create;
- ▶ An **object** is an instance of a particular class. E.g,

- ▶ A **method** is a member of a class that is a function. E.g.,

Note: The notes for last week were updated after the class to give a more coherent view on constructors and destructors. See <https://www.niallmadden.ie/2425-CS319/Week07/CS319-Week07.pdf>

.....

Before we continue with writing our own classes, we can now visit some important related topics in C++:

strings * **input and output streams** * **files**

3. Strings

A **string** is a collection of characters representing, for example, a word or a sentence.

In C++, a **char** array can be used to store a string. That approach is called a “C string”, since it is inherited from an older language, C.

Such “C strings” are not so easy to work with, so C++ provides its own **string** class. The class can be accessed once the **string** header file is included. It is part of the **std** namespace.

```
#include <string>
.
.
.
std::string name;
```

3. Strings

We have used `string` before (Weeks 1 and 2), but have not thought of it as a class.

Since it is a class, it has some methods, including:

- ▶ `length()` and `size()` which both return the number of characters in the string;
- ▶ `substr(i,l)` which returns a substring of length `l`, starting at position `i`.
- ▶ `find()` which finds the first occurrence of one substring in another.
- ▶ `c_str()` return the “C string” version. (Need this when working with files).

3. Strings

Example

Write a short C++ program that defines a `string` containing a sentence, and then extract the first word as another `string`.

00substring.cpp

```
2 #include <iostream>
  #include <string>

  int main(void)
6 {
    std::string
8     sentence="Ada Lovelace was the first programmer",
      first_word;
10  int space_loc = sentence.find(" ");    // Find first space
      first_word = sentence.substr(0,space_loc); // extract substring

      std::cout << "sentence is: " << sentence << std::endl;
14  std::cout << "first word is: '" << first_word << "'\n";
      return(0);
16 }
```

3. Strings

Expected output:

```
sentence is: Ada Lovelace was the first programmer  
first word is: 'Ada'
```

With numbers, we are used to working with special functions called **operators**, which are usually represented by a mathematical symbol, such as `+`, `-`, `=`, `*`, `/`, etc.

When writing our own **class**, we can overload some of these (more about the details later).

The **string** class overloads several operators:

- ▶ Assignment: `=`
- ▶ Relational: `==`, `>`, `<`, etc;
- ▶ Arithmetic: `+`, `+=`

01string-operators.cpp

```
2 #include <iostream>
#include <string>

int main(void)
6 {
    std::string name[3], // array of names
    long_name="";
    name[0]="Augusta";
10 name[1]="Ada";
    name[2]="King";

    long_name = name[0] + " " + name[1] + " " + name[2];

    std::cout << "long_name: " << long_name << std::endl;
16 return(0);
}
```

Output

```
1 long_name: Augusta Ada King
```

I/O means “Input/Output. So far, we have taken input from the keyboard, typically using `cin`, and sent output to a terminal window, using `cout`.

These are examples of **streams**: flows of data to or from your program. Moreover, they are examples of **objects** in C++.

In fact `cout` and `cin` are **objects** and are manipulated by their **methods**, i.e., public member functions and operators. (We saw this in Week 3)

Methods:

- ▶ `width(int x)` – minimum number of characters for next output,
- ▶ `fill(char x)` – character used to fill with in the case that the width needs to be elongated to fill the minimum.
- ▶ `precision(int x)` – sets the number of significant digits for floating-point numbers.

Code – width, fill

```
std::cout.fill('0');  
for (int i=0; i<8; i++)  
{  
    std::cout.width(6);  
    std::cout << rand()%200000  
               << std::endl;  
}
```

Output

```
089383  
130886  
092777  
036915  
147793  
038335  
085386  
160492
```

Code – precision

```
double Pi=3.1415926535;
for (int i=1; i<=8; i++)
{
    std::cout.precision(i);
    std::cout << "Pi (correct to "<< i << " digits) is "
                << Pi << std::endl;
}
```

Output

```
Pi (correct to 1 digits) is 3
Pi (correct to 2 digits) is 3.1
Pi (correct to 3 digits) is 3.14
Pi (correct to 4 digits) is 3.142
Pi (correct to 5 digits) is 3.1416
Pi (correct to 6 digits) is 3.14159
Pi (correct to 7 digits) is 3.141593
Pi (correct to 8 digits) is 3.1415927
```

- ▶ `setw` – like `width`
- ▶ `left` – Left justifies output in field width. Used after `setw(n)`.
- ▶ `right` – right justify.
- ▶ `endl` – inserts a newline into the stream and calls flush.
- ▶ `flush` – forces an output stream to write any buffered characters
- ▶ `dec` – changes the output format of number to be in decimal format
- ▶ `oct` – octal format
- ▶ `hex` – hexadecimal format
- ▶ `showpoint` – show the decimal point and some zeros with whole numbers

Others: `setprecision(n)`, `fixed`, `scientific`, `boolalpha`, `noboolalpha`, ...

Need to include `iomanip`

5. Files

All of the C++ programs we have looked at so far take their input from the *standard input stream*, which is usually the keyboard. Example:

```
std::cout << "Enter an integer: ";  
std::cin >> i;
```

Although the *standard input stream* can be redirected to be, for example, a file (easily done on a Mac and on Linux), it is usually necessary to open a file **from within the program** and take the data from there. The data is then processed and written to a new file.

5. Files

To achieve either of these tasks in **C++**, we create a **file stream** and use it just as we would `cin` or `cout`. We'll with a simple example.

02CountChars.cpp

- (i) This program opens an input file called `CPlusPlusTerms.txt`
- (ii) It opens an output file called `Output.txt`
- (iii) It counts the number of characters in the input file.
- (iv) It writes that result to the output file.

Download the input file from

<https://www.niallmadden.ie/2425-CS319>. Save it to the folder containing the executable that you compile.

Once we have the basic idea, we'll take a closer look at each operation (opening, reading, writing).

When working with files, we need to include the *fstream* header file.

To **read** from a file, declare an object of type *ifstream*.

Open the file by calling the *open()* method on that object.

02CountChars.cpp

```
8  #include <iostream>
   #include <fstream>
10 #include <cstdlib>

12 int main(void )
   {
14     char c;
       std::ifstream InFile;
16     InFile.open("CPlusPlusTerms.txt");
       std::cout << "Processing the contents of CPlusPlusTerms.txt"
18         << std::endl;
```

To **write** to a file, declare an object of type `ofstream`. Then (again) open the file by calling the `open()` method on that object.

To read a single character, can use `InFile.get()`

02CountChars.cpp

```
20  std::ofstream OutFile;  
    OutFile.open("Output.txt");  
22  std::cout << " See file Output.txt for more information."  
        << std::endl;  
  
    int i=0;  
26  InFile.get( c );
```

So "OutFile" is an object of type "ofstream"
and "ofstream" means "output file stream"

eof = "end of file"

If there are no more characters left in the input stream, then `InFile.eof()` evaluates as *true*.

Use the stream objects just as you would use `cin` or `cout`:

```
InFile >> data   or
OutFile << data.
```

Close the files:

```
InFile.close(),
OutFile.close()
```

01CountChars.cpp

```
28 while( ! InFile.eof() ) {
    i++;
    InFile.get( c );
30 }
32 OutFile << "CplusplusTerms.txt contains "
34         << i << " characters \n";
36 InFile.close();
    OutFile.close();
    return(0);
40 }
```

= "Stream insertion operator"

The method `open` works differently for `ifstream` and `ofstream`:

- ▶ `InFile.open()` Opens an existing file for reading,
- ▶ `OutFile.open()` Opens a file for writing. If it already exists, its contents are overwritten.

The first argument to `open()` contains the file name, and is an array of `char`acters. More precisely, it is of type `const char*`.

For example, we could have opened the input file in the last example with:

```
2  char InFileName[20]="CPlusPlusTerms.txt";  
3  ...  
4  std::cout << "Processing the contents of "  
5      << InFileName << std::endl;  
6  ...  
7  InFile.open(InFileName);
```

Note that file name is stored as a **"C string"**.

If we want to use C++ style strings, use the `c_str()` method. In this example we'll prompt the user to enter the file name.

```
2  std::ifstream InFile;  
   std::string InFileName;  
  
4  std::cout << "Input the name of a file: " << std::endl;  
   std::cin >> InFileName;  
  
   InFile.open(InFileName.c_str())
```

If you are typing the file name, there is a chance you will mis-type it, or have it placed in the wrong folder: so **always** check that the file was opened successfully. To do this, use the `fail()` function, which evaluates as `true` if the file was not opened correctly:

```
if (InFile.fail())
{
    std::cerr << "Error - cannot open " <<
        InFileName << std::endl;
    exit(1);
}
```

A better approach in this case might be to use a `while` loop, so the user can re-enter the filename. See [02CountCharsV02.cpp](#)

Recall that if you open an existing file for **output**, its contents are lost. If you wish to **append** data to the end of an existing file, use

To open an existing file and **append** to its contents, use

```
OutFile.open("Output.txt", std::ios::app);
```

.....

Other related functions include `is_open()` and, of course, `close()`

Above we also saw that `InFile.eof()` evaluates as *true* if we have reached the end of the (read) file.

Related to this are

```
InFile.clear(); // Clear the eof flag  
InFile.seekg(std::ios::beg); // rewind to beginning.
```

In the above example, we read a character from the file using `InFile.get(c)`. This reads the next character from the `InFile` stream and stores it in `c`. It will do this for any character, even non-printable ones (such as the newline char). For example, if we wanted to extend our code above to count the number of lines in the file, as well as the number of characters, we could use:

```
1  std::ifstream InFile;
   int CharCount=0, LineCount=0;
3  ...
   // Open the file, etc.
5  InFile.get( c );
   while( ! InFile.eof() ) {
7      CharCount++;
       if ( c == '\n' )
9          LineCount++;
       InFile.get( c );
11 }
```

— "if c is a newline character"

Alternatively, we could use the **stream extraction operator**:

```
InFile >> c;
```

However, this would ignore non-printable characters.

One can also use get() to read C-style strings. However, to achieve this task, it can be better to use getline(), which allows us to specify a delimiter character.

One of the complications of working with files, is knowing where to store input files so that your code can find them.

For some, IDEs, this is made additionally complicated by the fact that the compiled version of the program may not be in the same folder as the source code. So you have to work out where that is.

One way that can help, is change the `int main(void)` line to

```
1 int main(int argc, char * argv[])  
  {  
3     std::cout << "This program is running as " << argv[0];  
     std::cout << "\nDownload the input file to the same folder";  
5     std::cout << std::endl;
```

Alternatively, you can try opening a `ofstream` file with a very particular name, and then search for it.

If using an online compiler, you'll need one that allows multiple files, such as

<https://www.jdoodle.com/online-compiler-c++-ide>

6. Portable Bitmap Format (pbm)

Some self-study

We won't go through this section in class: please review in your own time.

Image analysis and processing is an important sub-field of scientific computing.

There are many different formats: you are probably familiar with JPEG/JPG, GIF, PNG, BMP, TIFF, and others. One of the simplest formats is the **Netpbm format**, which you can read about at https://en.wikipedia.org/wiki/Netpbm_format

6. Portable Bitmap Format (pbm)

There are three variants:

Portable BitMap files represent black-and-white images, and have file extension *.pbm*

Portable GrayMap files represent gray-scale images, and have file extension *.pgm*

Portable PixMap files represent 8-bit colour (RGB) images, and have file extension *.ppm*

In this example, we'll focus on *.pbm* files.

6. Portable Bitmap Format (pbm)

CS319.pbm

```
1 P1
2 25 9
3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4 0 1 1 1 1 0 1 1 1 1 0 1 1 1 1 0 0 1 0 0 1 1 1 1 0
5 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0
6 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0
7 0 1 0 0 0 0 1 1 1 1 0 1 1 1 1 0 0 1 0 0 1 1 1 1 0
8 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0
9 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0
10 0 1 1 1 1 0 1 1 1 1 0 1 1 1 1 0 1 1 0 0 0 0 0 0 0
11 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

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- ▶ The first line is the “magic number”. Here “P1” means that it is a PBM format ASCII (i.e, plain-text) file.
- ▶ The second line has two integer representing the number of columns and rows of pixels in the image, respectively.
- ▶ The remaining lines store the matrix of pixel values: 0 is “white”, and 1 is “black”.

6. Portable Bitmap Format (pbm)

The file `03FlipPBM.cpp` shows how to read such an image, and output its negative. (See notes from class).

03FlipPBM.cpp

```
16  std::ifstream InFile;
    std::ofstream OutFile;
    std::string InFileName, OutFileName;

20  std::cout << "Input the name of a PBM file: " << std::endl;
    std::cin >> InFileName;
    InFile.open(InFileName.c_str());
```

6. Portable Bitmap Format (pbm)

03FlipPBM.cpp

```
24 while (InFile.fail() )  
25 {  
26     std::cout << "Cannot open " << InFileName << " for reading."  
27     << std::endl;  
28     std::cout << "Enter another file name : ";  
29     std::cin >> InFileName;  
30     InFile.open(InFileName.c_str());  
31 }  
32 std::cout << "Successfully opened " << InFileName << std::endl;
```

6. Portable Bitmap Format (pbm)

03FlipPBM.cpp

```
34 // Open the output file
    OutFileName = "Negative_"+InFileName;
    OutFile.open(OutFileName.c_str());

    std::string line;
38 // Read the "P1" at the start of the file
    InFile >> line;
40 OutFile << "P1" << std::endl;

42 // Read the number of columns and rows
    unsigned int rows, cols;
44 InFile >> cols >> rows;
    OutFile << cols << " " << rows << std::endl;

    std::cout << "read: cols=" << cols << ", rows="
48         << rows << std::endl;
```

6. Portable Bitmap Format (pbm)

03FlipPBM.cpp

```
50  for (unsigned int i=0; i<rows; i++)
    {
52      for (unsigned int j=0; j<cols; j++)
          {
54          int pixel;
              InFile >> pixel;
56          OutFile << 1-pixel << " ";
              }
58      OutFile << std::endl;
    }
60  InFile.close();
    OutFile.close();

    std::cout << "Negative of " << InFileName << " written to "
64      << OutFileName << std::endl;
    return(0);
```

7. Review of classes

class

In C++, we defined new class with the `class` keyword.

An instance of the class is called an *"object"*.

A `class` combines by data and functions.

Within a class, code and data may be either

- ▶ **Private**: accessible only to another part of that object, or
- ▶ **Public**: other parts of the program can access it.

Roughly,

- ▶ keep data elements `private`,
- ▶ make function elements `public`.

Finish here
Friday

7. Review of classes

The basic syntax for defining a class:

```
class class-name {  
private:  
    ...    // private functions and variables  
public:  
    ...    // public functions and variables  
};
```

class-name becomes a new object type—one can now declare objects to be of type *class-name*.

This is only a declaration. Therefore,

- ▶ functions are not defined, though the prototype is given,
- ▶ variables are declared but are not initialised,
- ▶ the declaration block is delineated by { and }, and terminated with a semicolon.
- ▶ *scope resolution operator*, :: , used in function definition.

7. Review of classes

- ▶ A **Constructor** is a public method of a class, that has the same name as the class. It's return type is not specified explicitly. It is executed whenever a new instance of that class is created.
- ▶ A **destructor** is a method that is called on an object whenever it goes out of scope. The name of the destructor is the same as the class, but preceded by a tilde.

8. Vectors and Matrices

This is a course in Scientific Computing.

Many advanced and general problems in Scientific Computing are based around **vectors** and **matrices**. So one of our goals is to implement C++ classes for such structures, along with standard operations such as matrix-vector multiplication.

Along the way, we'll learn about

- ▶ operator overloading;
- ▶ **friend** functions and the **this** pointer;
- ▶ static variables.
- ▶ and much more

Our first step will be to study some problems and applications so that, before we design any classes or algorithms, we'll know what we will use them for. These problems include:

1. Basic analysis of matrices, for example with applications to image processing, graphs and networks.
2. Solution of linear systems of equations, for example with applications to data fitting;
3. Estimation of (certain) eigenvalues, for example with applications to Network Science.

Of these problems, probably the most ubiquitous is the solution of (large) systems of simultaneous equations.

That is, we want to solve a linear system of 3 equations in 3 unknowns: *find x_1, x_2, x_3 , such that*

$$3x_1 + 2x_2 + 4x_3 = 19$$

$$x_1 + 2x_2 + 3x_3 = 14$$

$$5x_1 + 1x_2 + 6x_3 = 25$$

This can be expressed as a **matrix-vector equation**:

More generally, the linear system of N equations in N unknowns:

find $x_1, x_2, \dots, x_N$, such that

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1N}x_N = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2N}x_N = b_2$$

$$\vdots$$

$$a_{N1}x_1 + a_{N2}x_2 + \cdots + a_{NN}x_N = b_N.$$

This, as a **matrix-vector equation** is:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \vdots & & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{pmatrix}$$

So, to proceed, we need to be able to represent **vectors** and **matrices** in our codes.

Our first focus will be on defining a class of vectors. Intuitively, we know it needs the following components:

Due to the level of detail in the matrix and vector classes, the following example is divided into three source files:

1. `Vector.h`, the header file which contains the class definition.
Include this header file in another source file with:
`#include "Vector.h"`
Note that this is **not** `<Vector.h>`
2. `Vector.cpp`, which includes the code for the methods in the `Vector` class;
3. `04TestVector.cpp`, a test stub.

In whatever compiler you are using, you'll need to create a **project** that contains all the files. (Ask Niall for help if needed).

See `Vector.h` for more details

```
2 // File: Vector.h (Version W08.1)
// Author: Niall Madden ¡Niall.Madden@UniversityOfGalway.ie¿
// Date: Week 8 of 2425-CS319
4 // What: Header file for vector class
// See also: Vector.cpp and 04TestVector.cpp
6 class Vector {
private:
8     double *entries;
    unsigned int N;
10 public:
    Vector(unsigned int Size=2);
12     ~Vector(void);

14     unsigned int size(void) {return N;};
    double geti(unsigned int i);
16     void seti(unsigned int i, double x);

18     void print(void);
    double norm(void); // Compute the 2-norm of a vector
20     void zero(void); // Set entries of vector to zero.
};
```

Vector.cpp

```
12 Vector::Vector(unsigned int Size)
13 {
14     N = Size;
15     entries = new double[Size];
16 }
17
18 Vector::~~Vector()
19 {
20     delete [] entries;
21 }
22
23 void Vector::seti(unsigned int i, double x)
24 {
25     if (i < N)
26         entries[i] = x;
27     else
28         std::cerr << "Vector::seti(): Index out of bounds."
29                     << std::endl;
30 }
```

Vector.cpp continued

```
32 double Vector::geti(unsigned int i)
   {
34     if (i<N)
        return(entries[i]);
36     else {
        std::cerr << "Vector::geti(): Index out of bounds."
38                << std::endl;
        return(0);
40     }
   }

void Vector::print(void)
44 {
    for (unsigned int i=0; i<N; i++)
46         std::cout << "[" << entries[i] << "]" << std::endl;
}
```

Vector.cpp continued

```
double Vector::norm(void)
50 {
    double x=0;
52     for (unsigned int i=0; i<N; i++)
        x+=entries[i]*entries[i];
54     return (sqrt(x));
}

void Vector::zero(void)
58 {
    for (unsigned int i=0; i<N; i++)
60         entries[i]=0;
}
```

Here is a simple implementation of a function that computes

$$c = \alpha a + \beta b$$

See 04TestVector.cpp for more details

```
14 // c = alpha*a + beta*b where a,b are vectors; alpha, beta are scalars
15 void VecAdd (vector &c, vector &a, vector &b,
16             double alpha, double beta)
17 {
18     unsigned int N;
19     N = a.size();
20
21     if ( (N != b.size()) )
22         std::cerr << "dimension mismatch in VecAdd " << std::endl;
23     else
24     {
25         for (unsigned int i=0; i<N; i++)
26             c.seti(i, alpha*a.geti(i)+beta*b.geti(i) );
27     }
28 }
```

10. Solving Linear Systems

We now move towards learning about **matrices**. When implementing the class, we will learn about

- ▶ operator overloading;
- ▶ **friend** functions and the **this** pointer;
- ▶ static variables.
- ▶ and much more

One of the most ubiquitous problems in scientific computing is the solution of (large) systems of simultaneous equations. That is, we want to solve a linear system of N equations in N unknowns: *find* $x_1, x_2, \dots, x_N$, *such that*

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1N}x_N = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2N}x_N = b_2$$

$$\vdots$$

$$a_{N1}x_1 + a_{N2}x_2 + \cdots + a_{NN}x_N = b_N.$$

There are several classic approaches:

1. Gaussian Elimination;
2. Related: *LU*- and Cholesky factorisation;
3. Stationary Iterative schemes such as **Jacobi's method**, **Gauss-Seidel** and Successive Over Relaxation (SOR);
4. Krylov subspace methods, of which Conjugate Gradients is the best known;
5. Enhancements of the Methods 3 and 4, using preconditioning with, for example, MultiGrid and Incomplete *LU*-factorisation.

Of the approaches listed above, Jacobi's is by far the simplest to implement, and so is the one we will study first.

See annotated slides.

